



Road Safety (Public Security)

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Data Cleanup

- Identify necessary features and drop unnecessary variables
- Proceeded to rename certain features to allow simplicity of use
- Due to most of the features being categorical, we replaced any missing/null values with NaN
- Realization: Binary features: null values are False
- Categorize non-binary feature values

```
import pandas as pd
import numpy as np

data = pd.read_csv('KSI.csv')
df = pd.DataFrame(data)

#dropping following columns
to_drop = ['X','Y','INDEX','ACCNUM','YEAR','TIME','STREET1','STREET2','OFFSET','WARDNUM','DIVISION','LATITUDE','LONGITUDE','ACCLOC','FATAL_NO','INITDIR','VEHTYPE','MANOEUVER','DRIVACT','DRIVCOND','PEDTYPE','PEDACT','PEDCOND','CYCLISTYPE','CYCACT','CYCCOND','PEDESTRIAN','CYCLIST','AUTOMOBILE','MOTORCYCLE','TRUCK','TRSN_CITY_VEH','EMERG_VEH','PASSENGER','POLICE_DIVISION','HOOD_ID','NEIGHBOURHOOD','ObjectId']

df.drop(to_drop, inplace=True, axis=1)

#missing values turning into NaN
missing_values=['?','na','X','<Null>','null',' ' ]
df = df.replace(missing_values,np.NaN)

#rename columns
new_names = {'ALCOHOL': 'isAlcohol',
'SPEEDING': 'isSpeeding',
'ROAD_CLASS': 'roadClass',
'RDSFCOND': 'roadCondition',
'AG_DRIV': 'isAggressive',
'DISABILITY': 'isDisable',
'INVAGE': 'ageInvolved',
'INVTYPE': 'involvementType',
'REDLIGHT': 'redlight',
'INJURY': 'injury'
}

df.rename(columns=new_names, inplace=True)

binarymap = {'Yes': int(True), 'No': int(False)}

df['isSpeeding'].fillna(int(False), inplace=True)
df['isSpeeding'] = df['isSpeeding'].replace(binarymap)

df['isAlcohol'].fillna(int(False), inplace=True)
df['isAlcohol'] = df['isAlcohol'].replace(binarymap)

df['redlight'].fillna(int(False), inplace=True)
df['redlight'] = df['redlight'].replace(binarymap)

df['isAggressive'].fillna(int(False), inplace=True)
df['isAggressive'] = df['isAggressive'].replace(binarymap)

df['injury'].fillna('None', inplace=True)
df['injury'] = df['injury'].replace({'None': 0, 'Minimal': 1, 'Minor': 2, 'Major': 3, 'Fatal': 4})

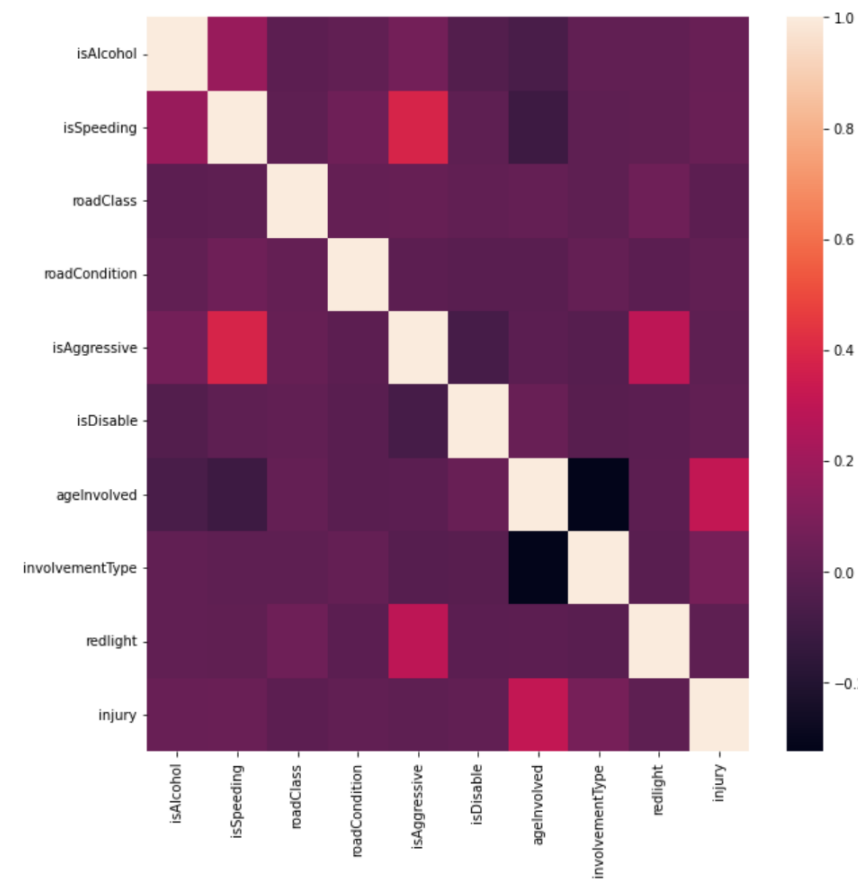
df['isDisable'].fillna(int(False), inplace=True)
df['isDisable'] = df['isDisable'].replace(binarymap)
```

Best Practices for Maintaining Data Hygiene



Data Exploration

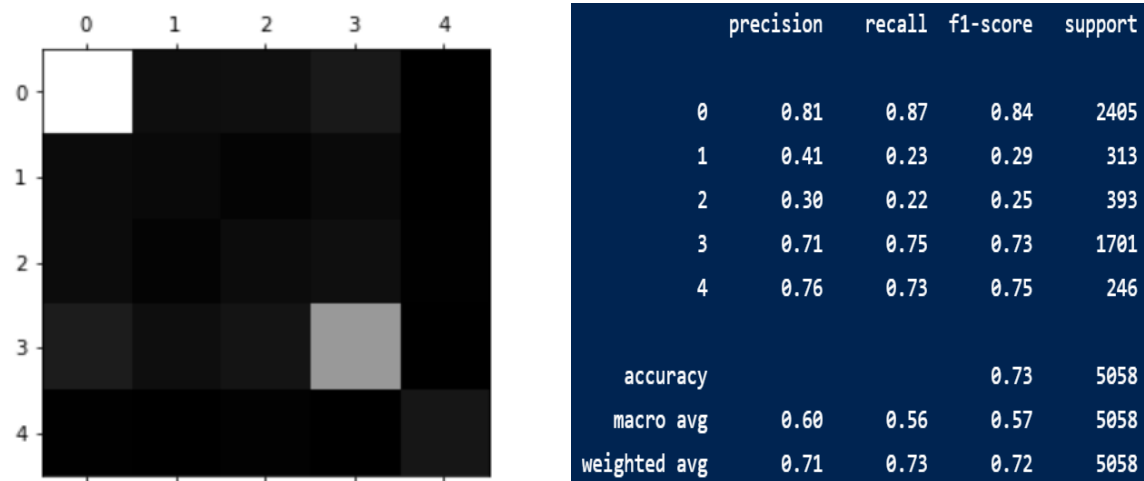
- Creating a correlation matrix for all features
- City Heat Map showing the severity of injury in each district
- Correlation amongst binary features. Relationships are observed amongst a few variables.



	HOUR	roadClass	LOCCOORD	TRAFFCTL	roadCondition	IMPACTYPE	involvementType	ageInvolved	injury	PEDESTRIAN	CYCLIST	AUTOMOBILE	MOTORCYCLE	TRUCK	TRSN_CITY_VEH	EMERG_VEH	PASSENGER	isSpeeding	isAggressive	redlight	isAlcohol	isDisable	HOOD_ID	MONTH	YEAR	WEEKDAY
HOUR	1.000000	0.015074	-0.004435	-0.060058	-0.028710	0.033343	-0.026011	0.044327	-0.019327	0.045341	0.042551	0.026569	0.073241	-0.048471	-0.035783	-0.005816	-0.011977	-0.086714	-0.033809	-0.064329	-0.093234	-0.009271	-0.011434	0.033214	0.027128	-0.056396
roadClass	0.015074	1.000000	-0.133036	0.135209	0.018042	0.002098	0.000137	0.020643	-0.007734	0.046172	-0.018510	-0.003672	-0.031826	0.001518	0.031061	-0.005095	-0.012406	-0.001931	0.022597	0.051708	-0.007877	0.009560	0.024319	0.011388	-0.010486	0.014050
LOCCOORD	-0.004435	-0.133036	1.000000	-0.622242	0.020853	0.017292	0.012162	-0.045413	0.017725	-0.052693	-0.040831	-0.024943	0.024695	0.001735	-0.020264	-0.014551	0.016689	0.109824	-0.130971	-0.196580	0.062650	0.031787	-0.019478	0.003708	-0.044425	0.020294
TRAFFCTL	-0.060058	0.135209	-0.622242	1.000000	-0.036434	-0.022465	-0.005811	0.054663	-0.012794	0.021731	0.005972	0.028870	-0.049014	0.033640	0.002422	0.007822	0.035046	-0.111506	0.193240	0.337144	-0.053579	-0.020322	0.003190	-0.002664	0.023764	-0.006117
roadCondition	-0.028710	0.018042	0.020853	-0.036434	1.000000	-0.036178	0.019483	-0.019831	0.007101	0.028064	-0.074698	0.001296	-0.082454	-0.009578	0.030935	0.003464	0.002858	0.047805	-0.008414	-0.009973	0.007911	-0.015772	-0.010590	-0.092492	-0.053734	0.017011
IMPACTYPE	0.033343	0.002098	0.017292	-0.022465	-0.026178	1.000000	0.029301	-0.051770	0.017906	-0.168646	-0.307587	0.038784	0.280604	-0.024290	-0.048409	-0.049301	0.105118	0.074140	0.011907	-0.214679	0.012272	0.033445	-0.016833	0.013961	0.006249	0.009638
involvementType	-0.026011	0.000137	0.012162	-0.005811	0.019483	0.029301	1.000000	-0.324816	0.076239	0.200507	-0.228897	-0.136916	-0.040050	0.190226	0.049834	-0.001077	0.088434	-0.001579	-0.027215	-0.016045	0.010327	-0.024480	-0.001936	-0.011095	-0.112667	-0.006283
ageInvolved	0.044327	0.020643	-0.045413	0.054663	-0.015831	-0.051770	-0.324816	1.000000	0.307255	0.117012	-0.001830	0.019047	-0.020624	-0.025143	-0.011425	0.015547	-0.085002	-0.110336	-0.013740	-0.005066	-0.069406	0.030965	0.003925	0.015569	0.166226	-0.037781
injury	-0.019327	-0.007734	0.017725	-0.012794	0.007101	0.017906	0.076239	0.307255	1.000000	-0.027440	-0.018041	-0.019569	0.022032	-0.024692	-0.025716	-0.012286	-0.043893	0.034767	0.000740	0.001067	0.030983	0.009371	-0.003902	0.002601	0.059664	0.007701
PEDESTRIAN	0.045341	0.046172	-0.052693	0.021731	0.028064	-0.168646	0.200507	0.117012	-0.027440	1.000000	-0.269551	-0.108482	-0.219831	-0.021541	0.021692	-0.018481	-0.354996	-0.207654	-0.164907	-0.141696	-0.088672	-0.102606	0.047769	0.038751	0.393327	-0.068103
CYCLIST	0.042551	-0.016510	-0.046831	0.005972	-0.074698	-0.307587	-0.228897	-0.001830	-0.018041	-0.269551	1.000000	-0.074286	-0.097083	0.010465	-0.023319	-0.007792	-0.165409	-0.106098	-0.106744	-0.064650	-0.053732	0.058034	0.033267	0.054747	0.032835	-0.023974
AUTOMOBILE	0.026569	-0.003672	-0.024943	0.028870	0.001296	0.038784	-0.136916	0.019047	-0.019569	-0.108482	-0.074286	1.000000	-0.109569	-0.371544	-0.435462	-0.024646	0.133595	0.064337	0.144973	0.090222	0.061158	0.052553	-0.025869	-0.008882	-0.029352	0.049848
MOTORCYCLE	0.073241	-0.031826	0.024695	-0.049014	-0.082454	0.280604	-0.040050	-0.020624	0.022032	-0.219831	-0.097083	-0.109569	1.000000	-0.047141	-0.060057	-0.012835	-0.073214	0.074761	0.067706	-0.028625	-0.037641	-0.040007	0.009204	0.036679	0.009910	0.030997
TRUCK	-0.048471	0.001518	0.001735	0.033640	-0.009578	-0.024290	0.190226	-0.025143	-0.024692	-0.021541	0.010465	-0.371544	-0.047141	1.000000	-0.016063	-0.011016	-0.028046	0.002720	-0.013245	-0.013422	-0.036904	-0.014880	-0.056197	0.012785	-0.003645	-0.048439
TRSN_CITY_VEH	-0.035783	0.031061	-0.020264	0.002422	0.030935	-0.048409	0.049834	-0.011425	-0.025716	0.021692	-0.023319	-0.435462	-0.060057	-0.016063	1.000000	-0.010840	0.017269	-0.018990	-0.070421	-0.001345	-0.015026	-0.035072	0.000455	-0.021990	0.013277	-0.019879
EMERG_VEH	-0.005816	-0.005095	-0.014551	0.008722	0.003464	-0.049301	-0.001077	0.015547	-0.012286	-0.018481	0.007792	-0.024646	-0.012835	-0.011016	-0.010840	1.000000	0.029426	0.011320	0.019388	0.017056	-0.009189	0.034886	-0.009967	0.017734	0.036156	-0.000785
PASSENGER	-0.011977	-0.012406	0.016689	0.035046	0.002858	0.051118	0.088434	-0.089002	-0.043893	-0.354996	-0.165409	0.133595	-0.073214	-0.028046	0.017269	0.029426	1.000000	0.171512	0.130504	0.138095	0.087949	0.012963	-0.022005	-0.045667	0.404088	0.009911
isSpeeding	-0.086714	-0.001931	0.109824	-0.111506	0.047805	0.074140	-0.001579	-0.110336	0.034767	-0.207654	-0.106098	0.064337	0.074761	0.002720	-0.018990	0.011320	0.171512	1.000000	0.383327	0.004409	0.180159	-0.000450	-0.022740	-0.040522	0.026553	0.068779
isAggressive	-0.033809	-0.001931	-0.130971	0.193240	-0.008414	0.011907	-0.027215	-0.013740	0.000740	-0.164907	-0.106744	0.144973	0.067706	-0.013245	-0.070421	0.019388	0.130504	0.383327	1.000000	0.292353	0.067560	-0.027327	-0.044147	-0.033143	0.020471	-0.004417
redlight	-0.064329	0.051708	-0.196580	0.337144	-0.009973	-0.214679	-0.016045	-0.005066	0.001067	-0.141696	-0.064650	0.090222	-0.028625	-0.013422	-0.001345	0.017056	0.138095	0.004409	0.292353	1.000000	0.010722	-0.010582	-0.003893	-0.026976	-0.023278	0.026277
isAlcohol	-0.093234	-0.007877	0.062650	-0.053579	0.007911	0.012272	0.010327	-0.069406	0.030983	-0.088672	-0.053732	0.061158	-0.037641	-0.036904	-0.015026	-0.009189	0.087949	0.180159	0.067560	0.010722	1.000000	-0.036177	-0.003713	-0.017545	-0.047574	0.078523
isDisable	-0.009271	0.009560	0.031787	-0.020322	-0.015772	0.033445	-0.024480	0.003925	0.009371	-0.102606	-0.058034	0.052553	-0.040007	-0.014880	-0.03072	-0.009886	0.012963	-0.000455	-0.078444	-0.036177	1.000000	0.000746	-0.013244	-0.023762	-0.024139	-0.001617
HOOD_ID	-0.011434	0.024319	-0.019478	0.003190	-0.010590	-0.016833	-0.001936	0.003925	-0.003902	0.047769	0.033267	-0.025869	0.009204	-0.056197	0.000455	-0.009967	-0.022005	-0.022740	-0.073237	-0.030893	-0.003713	0.006746	1.000000	0.001637	-0.018037	0.020914
MONTH	0.033214	0.011388	0.003708	-0.002664	-0.092492	0.013961	-0.011095	0.015569	0.002601	0.038751	0.045747	-0.008882	0.036679	0.012785	-0.021990	0.017734	-0.045667	-0.040522	-0.044147	-0.026976	-0.017545	-0.013244	0.007637	1.000000	0.005037	0.060646
YEAR	0.027128	-0.010486	-0.044425	0.023764	-0.053734	0.006249	-0.112667	0.166226	0.059664	0.039327	0.028355	-0.029352	0.099101	-0.003645	0.013277	0.036156	0.040088	0.026553	0.033143	-0.029278	-0.047574	-0.023762	0.018037	0.005037	1.000000	0.015551
WEEKDAY	-0.056396	0.014050	0.020294	-0.006117	0.017011	0.009638	-0.037781	0.007701	-0.068103	-0.023974	0.049848	0.003997	-0.084349	-0.019879	-0.000785	0.009911	0.068779	0.020471	0.026277	0.078523	-0.024139	0.020914	0.006046	0.015551	1.000000	

Models

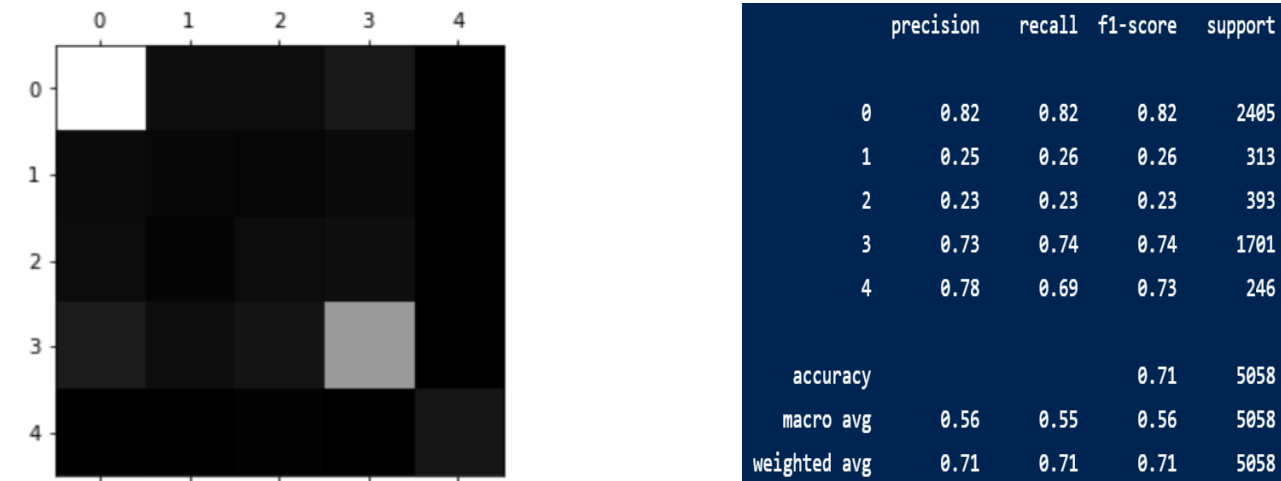
Random Forest



Observations

- Observed to have the highest accuracy score when comparing against other models
- Might be the best option as we do not have to concern ourselves with underfitting/overfitting
- High dimensional data: Ideal as working with many subsets
- Might be the ideal model as due to its versatility, low bias (though high variance) and fast prediction speeds

Decision Tree



Observations

- Observed to have the highest accuracy score when comparing against other models
- Might be the best option as we do not have to concern ourselves with underfitting/overfitting
- High dimensional data: Ideal as working with many subsets
- Might be the ideal model as due to its versatility, low bias (though high variance) and fast prediction speeds

Models

Model	Accuracy Score	Observations
AdaBoost Classifier	0.65	• Low/Mid accuracy score
SVM classifier	0.48	• Low accuracy score
KNN	0.62	• Low/Mid accuracy score
Logistic Regression	0.56	• Lower accuracy score
SGD Classifier	0.33	• Very low accuracy score
Naive Bayes	0.48	• Low accuracy score
Voting Classifier	0.72	• Higher accuracy score than majority of models

Fine Tuning – Decision Tree

Model	Accuracy Score	Observations
AdaBoost Classifier	0.65	
SVM classifier	0.48	
KNN	0.62	
Logistic Regression	0.56	
SGD Classifier	0.33	
Naive Bayes	0.48	
Voting Classifier	0.72	

Conclusion

Step	Observations
Data Cleanup	
Data Analysis	
Data Exploration	
Models	
Hyperparameters	

In the real world, this data analysis and modelling would be helpful because....